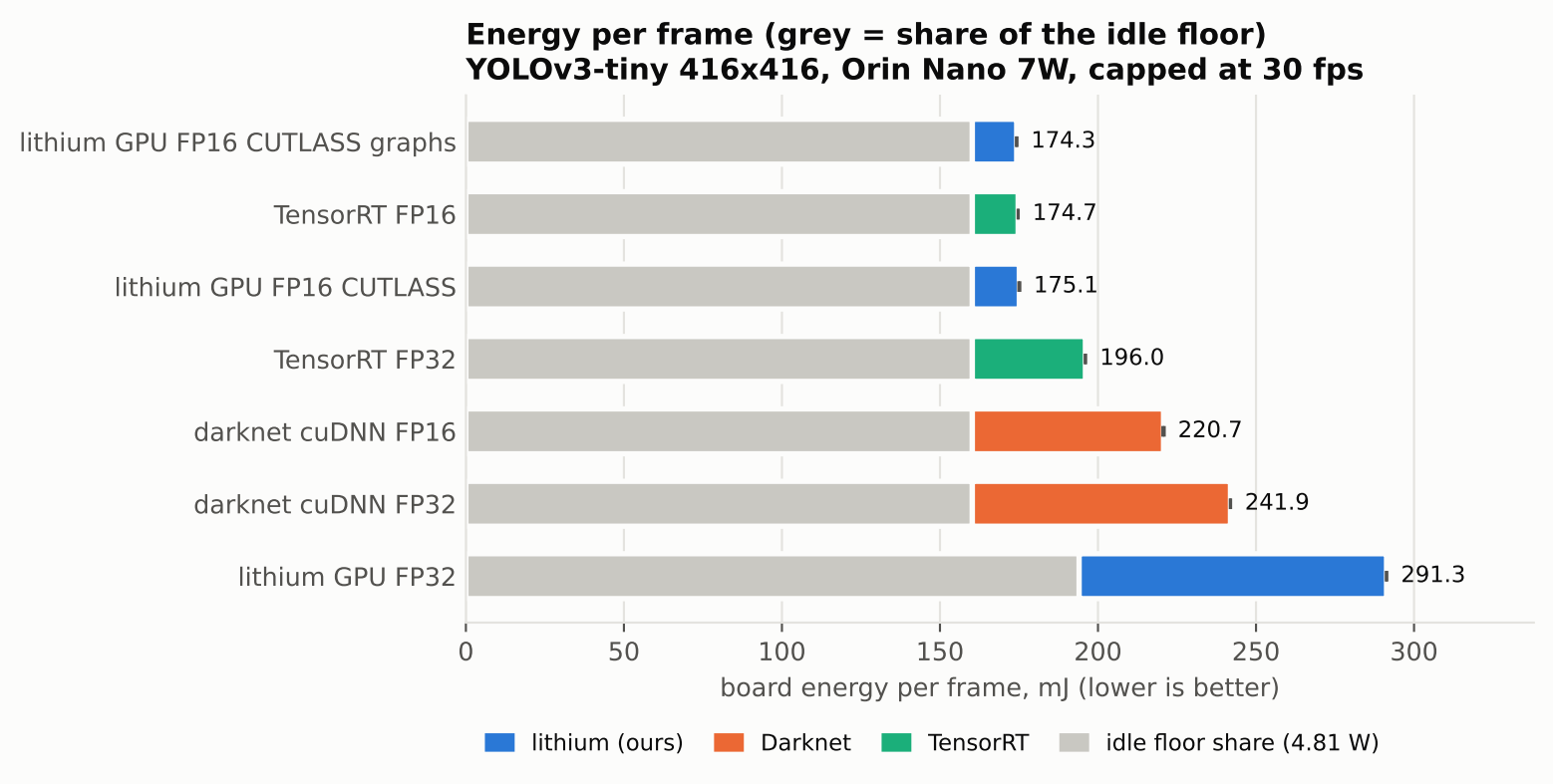
Energy per frame (grey = share of the idle floor)
YOLOv3-tiny 416x416, Orin Nano 7W, capped at 30 fps
lithium GPU FP16 CUTLASS graphs
TensorRT FP16
lithium GPU FP16 CUTLASS
TensorRT FP32
darknet cuDNN FP16
darknet cuDNN FP32
lithium GPU FP32
174.3
174.7
175.1
196.0
220.7
241.9
291.3
0
50
100
150
200
250
300
board energy per frame, mJ (lower is better)
lithium (ours)
Darknet
TensorRT
idle floor share (4.81 W)